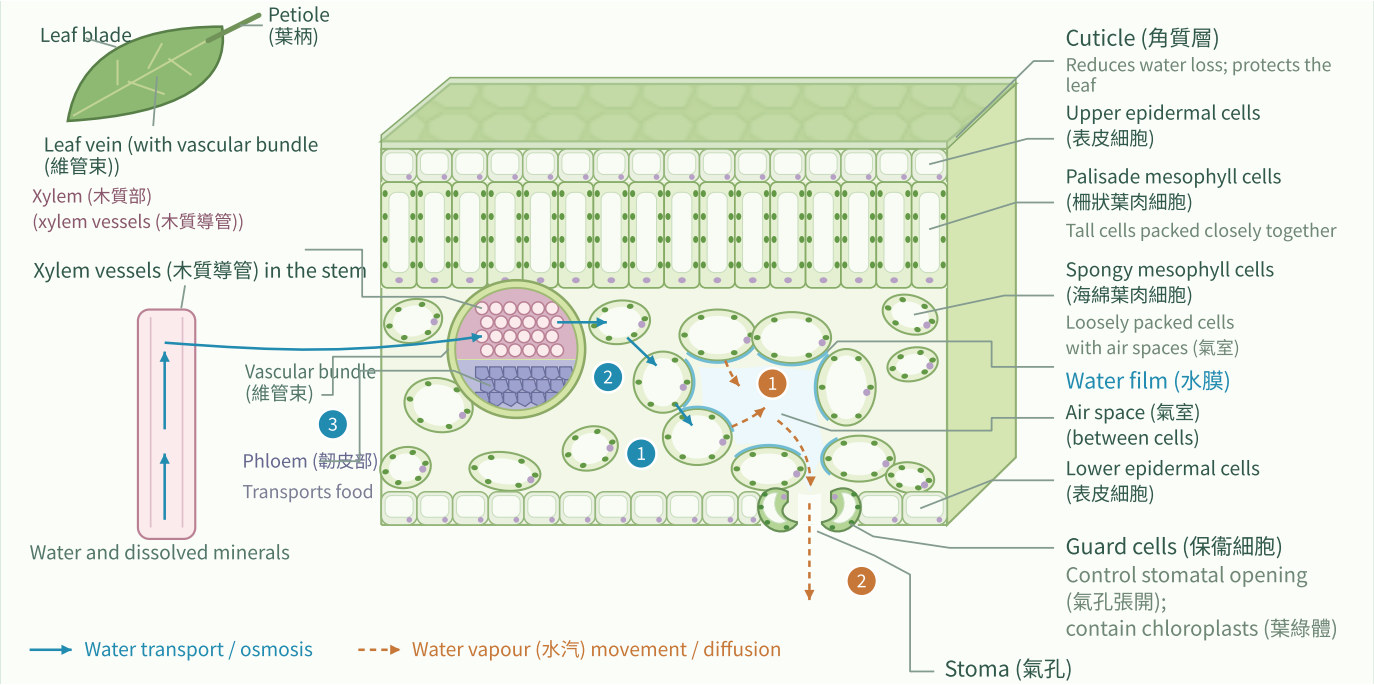
Name: _____ Class and student number: _____

Red water traces and water transport in celery

Read A from the start and while doing B. Receive C only after completing and keeping answers in B.

Read first: water transport and transpiration (蒸騰) in a leaf



Transpiration (蒸騰) through stomata (氣孔)

1 Water evaporates (蒸發) from the surfaces of mesophyll cells (葉肉細胞) into the air spaces (氣室).

2 Water vapour (水汽) in the air spaces (氣室) diffuses through stomata (氣孔) into the air.

How transpiration pull (蒸騰拉力) forms

1 Mesophyll cells (葉肉細胞) lose water, so their water potential falls.

2 Mesophyll cells (葉肉細胞) take in water from nearby cells by osmosis.

3 Water is drawn out of the xylem vessels (木質導管), creating transpiration pull (蒸騰拉力).

1. Transpiration (蒸騰) is the process in which water in a plant evaporates (蒸發) and is lost to the air as water vapour (水汽). Water vapour (水汽) mainly leaves through stomata (氣孔) in leaves and herbaceous stems (草本莖).
   a. Water evaporates (蒸發) from the water film (水膜) on the surface of mesophyll cells (葉肉細胞) and enters the air spaces (氣室) in the spongy mesophyll as water vapour (水汽). When the water vapour (水汽) concentration in the air spaces (氣室) is higher than in the air outside, water vapour (水汽) leaves the leaf through stomata (氣孔) by diffusion.
   b. When mesophyll cells (葉肉細胞) lose water, their water potential falls. They take in water from nearby cells by osmosis. This draws water out of the leaf's xylem vessels (木質導管) and creates transpiration pull (蒸騰拉力), which pulls water upwards through the stem's xylem vessels (木質導管).
2. Under suitable conditions, such as enough water being available, higher light intensity (光強度) usually makes stomata (氣孔) open wider. Water vapour (水汽) then leaves the leaf faster by diffusion, so the transpiration rate (蒸騰速率) increases.
   Note: More light may also make the leaf's temperature rise, increasing transpiration (蒸騰) further. This investigation therefore assumes that all setups have the same temperature so we can compare the effect of light intensity (光強度).
3. Transpiration pull (蒸騰拉力) keeps roots taking in water and moves water and dissolved minerals absorbed by the roots upwards through xylem vessels (木質導管) to the leaves.

See textbook section 11.1, pages 11-3 to 11-8. The red water trace gives evidence of water transport. Its upward speed is not a direct measure of the transpiration rate (蒸騰速率). Comparing celery with and without leaves alone also cannot prove the role of stomata (氣孔).

Internal leaf structure and transpiration (蒸騰)

Orange numbers: transpiration (蒸騰) through stomata (氣孔) | Blue numbers: how transpiration pull (蒸騰拉力) forms

Enlarged view, not to scale. The celery stalk in this experiment is a petiole (葉柄). The stem shown here explains transport through xylem vessels (木質導管).

Worked methods (not answers for A–D)

Explanation → worked example → supported practice → independent application → checking and correction

Distinguish processes and plant parts

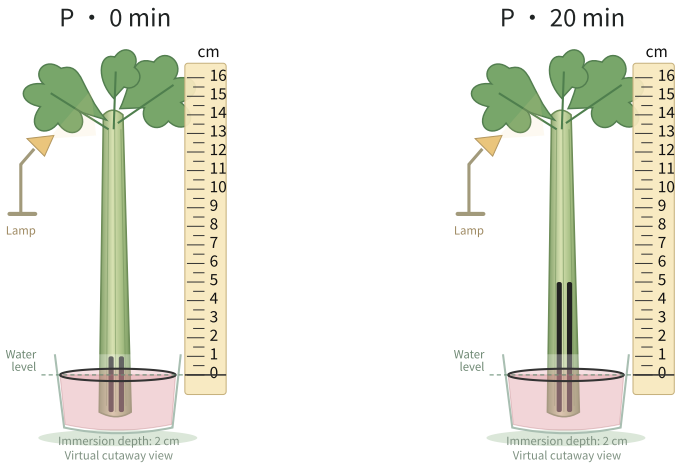
Water transport is liquid water moving through xylem (木質部). Transpiration (蒸騰) is water loss from a plant to the air. The trace gives evidence of liquid water transport, not a direct measurement of water vapour (水汽) lost. The long celery stalk is a petiole (葉柄) containing xylem (木質部), not a stem.

Hypothesis and fair-comparison example

A testable (可測試的) hypothesis states the factor changed on purpose and a measurable result. Example: “With other conditions the same, celery with a larger leaf area has a faster average upward speed of the red trace.” Compare two leaf areas, keep light and other conditions the same, and measure in cm/min. Results may not support the hypothesis.

Independent variable: the factor changed on purpose. Dependent variable: the measured result. Controlled variables: factors kept the same. A fair comparison changes only the factor being studied. When comparing leaf area, keep light, temperature and measurement time the same.

Ruler and calculation example



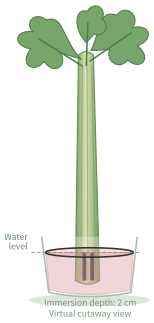
Neutral practice sample P (a separate method example): the left image is 1.0 cm at 0 min; the right image is 5.0 cm at 20 min. Read the ruler at the level of the trace top. The liquid surface is the fixed zero; each small division is 0.5 cm.

Example average speed = (5.0 − 1.0) ÷ 20 = 0.20 cm/min. Do not simply divide final height by time; subtract the starting reading first.

Plotting example

Another method example: light intensity (光強度) 500 lux, speed 0.20 cm/min. Find 500 on X and 0.20 on Y; mark their intersection. These values teach the method only, not investigation setup data.

Supported practice





After some time

What the school gardening group found

The gardening group puts celery with leaves in red water. They observe it before and after some time has passed.

Look at the two pictures on the left and record the changes you see. The group wants to know: do light and leaves affect water transport?

Main investigation question

When other conditions are the same, how does light intensity (光強度) affect the upward speed of the red water trace in celery with leaves?

This diagram shows an imaginary cutaway so you can see the water trace inside. The celery used in the experiment has not been cut lengthwise.

Your first observations

Describe visible changes in the trace; keep explanations separate from observations.

Evaluate a hypothesis, not original prediction

Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.

Identify the variables

For the light comparison using A, B and C, tick one column for each factor.

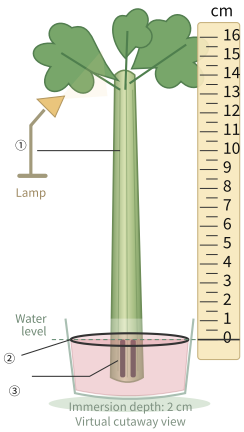
Factor	Independent variable	Dependent variable	Controlled variables
Light intensity (光強度)	☐	☐	☐
Height / average upward speed of the red water trace	☐	☐	☐
Temperature	☐	☐	☐
Humidity (濕度) and air movement	☐	☐	☐
Red water concentration and depth of immersion (浸入深度)	☐	☐	☐
Total leaf area of the setups with leaves	☐	☐	☐
Length and thickness of the celery stalks	☐	☐	☐

Read specified setups; do not design from scratch

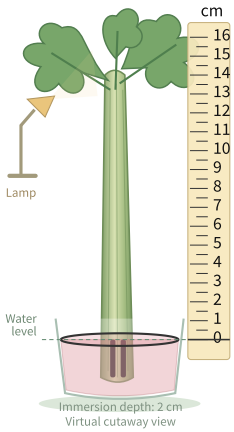
Fixed conditions: 25°C, the same humidity (濕度) and airflow, dye concentration, 2 cm depth of immersion (浸入深度), and similar petiole (葉柄) length and thickness. A, B and C have the same leaf area. B and D have the same relevant conditions except leaf presence.

0 min

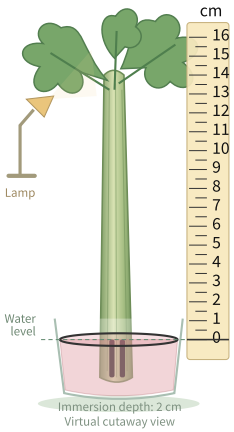
Setup A • With leaves
10 cm • 1200 lux



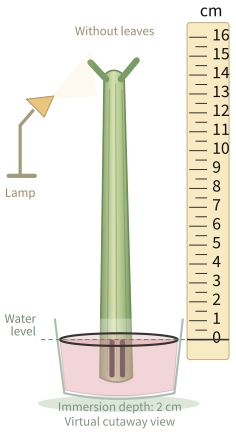
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



① points to the long stalk, ② to the liquid surface, ③ to the immersed part. Complete: plant part and water-carrying tissue for ①; ruler reading at ②; immersion depth for ③.

For light intensity (光強度) use (A/B/C / A/D); for leaf presence use (B/D / A/D). Why is A/D not a fair comparison of leaf presence?

What assumptions does this investigation make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all the assumptions that apply. You may choose more than one.

- ☐ Changing the lamp distance does not make the setups have different temperatures.
- ☐ Setups A, B and C have the same total leaf area.
- ☐ Leaf area does not affect the results, so it does not need to be kept the same.
- ☐ Humidity (濕度) and air movement are the same for all setups.

Changing lamp distance may change temperature. Why monitor and keep all setups at 25°C?

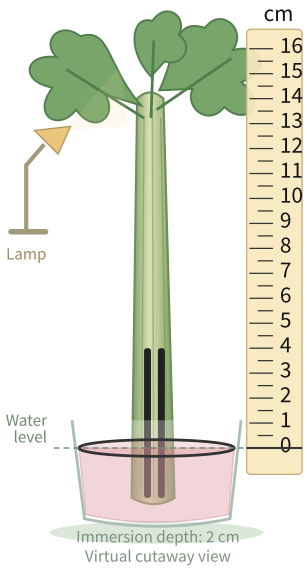
Independent application: read, calculate and analyse

Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.

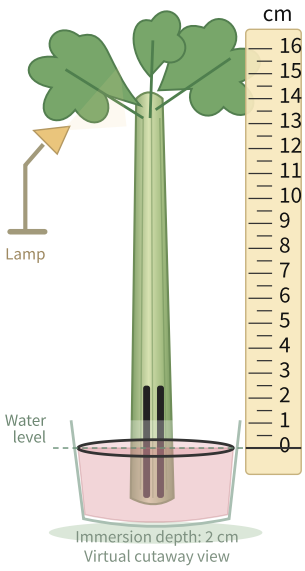
Images use the same teaching model as the VL, not real measurements. Zero stays at the liquid surface; each small division is 0.5 cm. Read the rulers yourself; no answers are listed beside them.

10 min

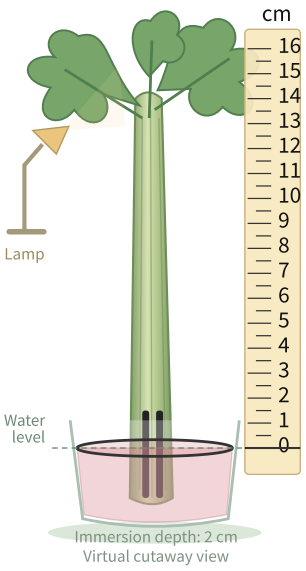
Setup A • With leaves
10 cm • 1200 lux



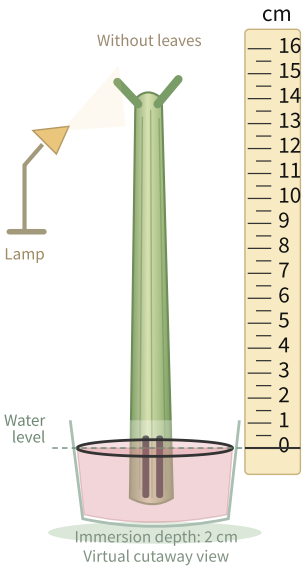
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux

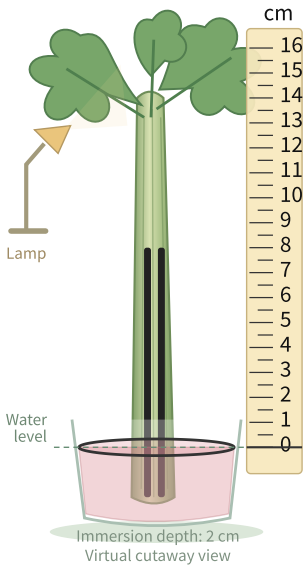


Setup D • Without leaves
20 cm • 300 lux

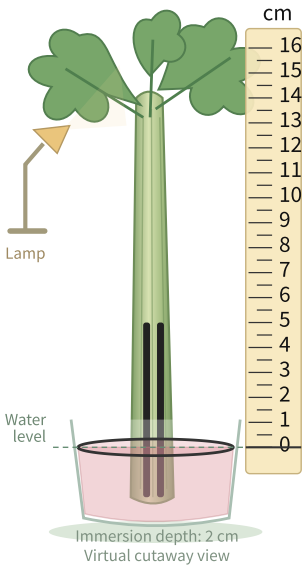


20 min

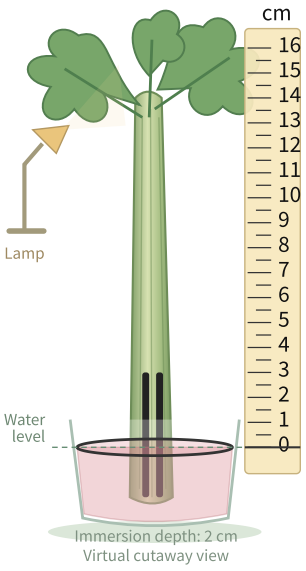
Setup A • With leaves
10 cm • 1200 lux



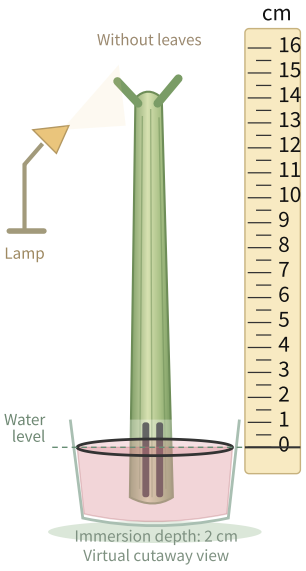
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



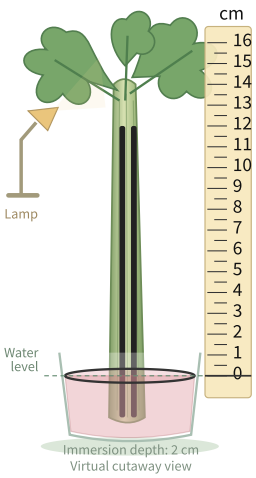
Setup D • Without leaves
20 cm • 300 lux



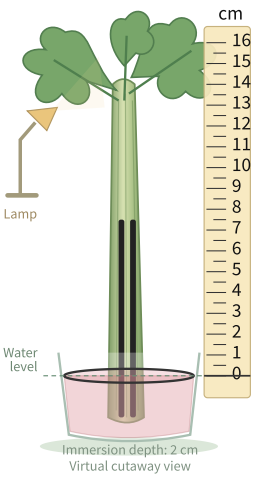
Keep first answers; write corrections in C after checking, without erasing first answers.

30 min

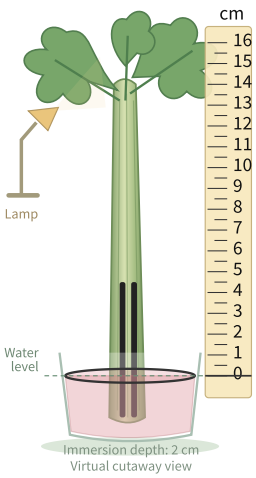
Setup A • With leaves
10 cm • 1200 lux



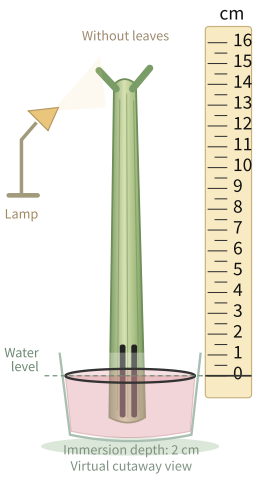
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Your measurement record

min	A (cm)	B (cm)	C (cm)	D (cm)
0				
10				
20				
30				

Keep first answers; write corrections in C after checking, without erasing first answers.

Calculate the average upward speed

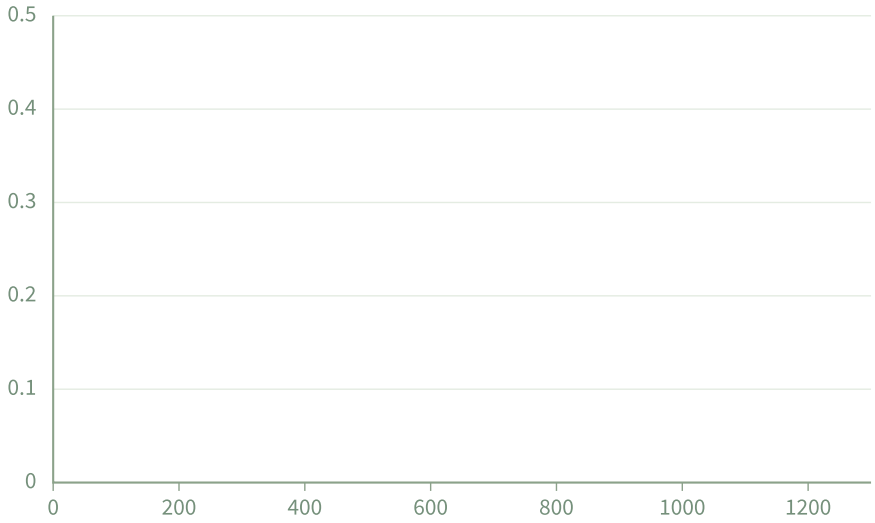
Average upward speed = (height at 30 min − height at 0 min) ÷ 30. Use two decimal places and cm/min.

Setup	Average upward speed (cm/min)
A	
B	
C	
D	

Light intensity (光強度) and the upward speed of the red water trace

Plot four points using your calculations. X: light intensity (光強度) (lux); Y: average upward speed (cm/min). Join A, B and C in light-intensity order to show a reasonable trend. Mark D separately; do not join it into that trend.

Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

Use your graph to reach a conclusion

Circle one answer in each set of brackets.

1. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend).

2. What is the role of leaves?

At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar).

This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards).

3. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).)

Use data from A, B, C and B/D to support the conclusions.

Can three light-intensity points give exact speeds for all untested values? (yes / no)

Keep first answers; write corrections in C after checking, without erasing first answers.

Name: _____ Class and student number: _____

Check answers

Read this booklet only after completing and keeping first answers in B. Use a different pen colour to check and correct; do not erase first answers.

The trace is higher in the later image: a clue to upward transport, not a direct view of water vapour (水汽) lost.

No: “better” has no defined measure. The factor is leaf area; replace “better” with faster average upward speed (cm/min), keeping other conditions the same. Testable (可測試的) does not mean proved.

① Petiole (葉柄), xylem (木質部); ② 0 cm; ③ 2 cm. Use A/B/C for light, B/D for leaves. A/D changes light and leaf presence together and cannot isolate the leaf factor.

Group	Factors
Independent variable	Light intensity (光強度)
Dependent variable	Height / average upward speed of the red water trace
Controlled variables	Temperature / Humidity (濕度) and air movement / Red water concentration and depth of immersion (浸入深度) / Total leaf area of the setups with leaves / Length and thickness of the celery stalks

What assumptions does this investigation make?

Changing the lamp distance does not make the setups have different temperatures. / Setups A, B and C have the same total leaf area. / Humidity (濕度) and air movement are the same for all setups.

Monitor and maintain 25°C so lamp heating does not become another factor affecting transpiration (蒸騰) and transport. Also control leaf area, water supply, humidity (濕度) and airflow; do not merely assume no heating.

Model readings and mean speeds

min	A (cm)	B (cm)	C (cm)	D (cm)
0	0.0	0.0	0.0	0.0
10	4.0	2.5	1.5	0.5
20	8.0	5.0	3.0	1.0
30	12.0	7.5	4.5	1.5

Setup	Average upward speed (cm/min)
A	0.40
B	0.25
C	0.15
D	0.05

1. How is light intensity (光強度) related to the upward speed of the water trace? increase

2. What is the role of leaves? faster

2. What is the role of leaves? help water move upwards

3. What can this experiment tell us? The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率).

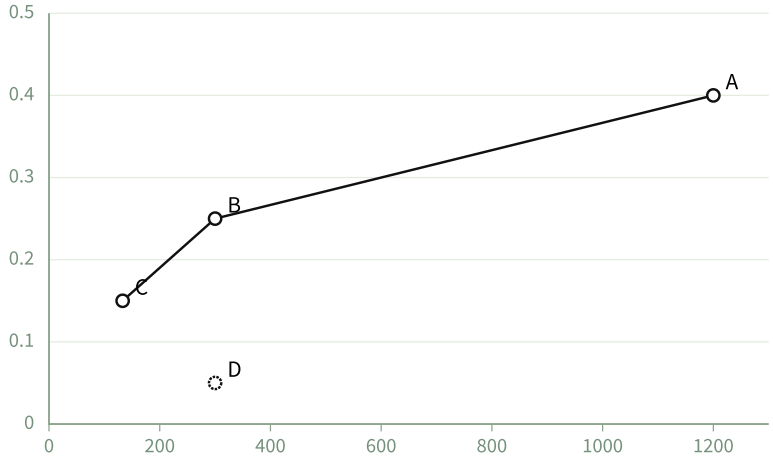
Trace speed is indirect evidence of transport, not a direct transpiration rate (蒸騰速率); water vapour (水汽) loss was not measured. Leaves/no leaves alone cannot prove that water vapour (水汽) escapes only through stomata (氣孔).

Untested light intensities cannot be predicted exactly.

Check the graph, explain and correct

Read this booklet only after completing and keeping first answers in B. Use a different pen colour to check and correct; do not erase first answers.

Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

The model gives 0.15, 0.25 and 0.40 cm/min for the group with leaves at 133, 300 and 1200 lux. At 300 lux, B is 0.25 and D is 0.05 cm/min, supporting a leaf role in upward transport. Three points do not establish an exact untested curve.

Use A/B/C and B/D data to explain light, transpiration (蒸騰) and xylem (木質部) transport, including suitable conditions and evidence limits.

Choose a wrong or incomplete answer of your own: state it and correct it with data or concepts. If none, correct “trace speed is transpiration rate (蒸騰速率)”.

Explanation reference (try your own explanation first):

With suitable conditions and enough water, stronger light usually promotes stomatal opening (氣孔張開) and transpiration (蒸騰). Water evaporates (蒸發) from mesophyll surfaces; vapour diffuses through stomata (氣孔). Water potential falls, water is replaced by osmosis and pulled from xylem (木質部), forming transpiration pull (蒸騰拉力). Data support a transport trend, not a direct transpiration rate (蒸騰速率).

State the original error and correct it using data or concepts. Example: replace “trace speed equals transpiration rate (蒸騰速率)” with “a transport clue; water vapour (水汽) loss needs a separate measurement”. Accept other reasonable corrections.

For teachers only. Do not give this part to students.

Red water traces in celery: does light change water transport?

Distribution, clarification, feedback and timing

1. Study conditions: experimental group uses IBL plus VL; comparison group has no prior lecture, does not operate VL, and learns from A, B and C alone. Allow 15–20 minutes per group, targeting 18 minutes for both; pilot actual time.
2. Minutes 0–4: give out A and B together. Students read concepts, the leaf diagram and worked examples in A themselves. Core teaching is available from the start.
3. Minutes 4–7: B1–B2 supported practice, specified setups and fair comparisons. Students may consult A; no design from scratch.
4. Minutes 7–15: B3–B5 ruler reading, calculation, plotting and analysis. Save first B answers before giving C; record unfinished work within the common time limit, not automatic zero.
5. Minutes 15–18: check with C, explain using data and correct errors. A remains available. D and the full pack are teacher-only; do not distribute C or D at the start.
6. Teachers clarify procedures such as page order, circling, units or ruler position only. No prior lecture or answer hints before B is completed. Record clarifications and actual time; use the standard C key for feedback.
7. Both groups share concepts, skill goals and model data with similar time; teaching order and interaction differ. Hypothesis evaluation after teaching is not original hypothesis performance. C provides the full key; D supplies alignment and feedback.
8. Use separate common tests such as proposal SPSAI / CKT for effectiveness, not activity totals. Researchers must confirm test versions, language, raters, feasibility within 15–20 minutes, and grayscale printing.

Paper ruler reading is not full experimental operation. Do not directly compare paper scores with VL original prediction, operation or completion scores; evaluate effectiveness through separate common tests.

This guide is a draft for piloting and rater calibration, not a validated instrument.

For teachers only. Do not give this part to students.

Feedback by area (not a group-effectiveness total)

Record areas separately, without a total compared with the VL 32-point score. Missing responses are unfinished / awaiting assessment. Paper reading is not full practical operation. Use separate common tests for effectiveness.

Area	2: sufficient	1: partial	0: needs correction
Observation	Describes change and separates inference	Some visible change	Observation needs correction
Classification	All seven factors correct	Some correct	Classification needs correction
Understanding inquiry	Identifies testable (可測試的) factor, outcome and fair comparison reasons	Incomplete reasons	Confuses factors / comparisons
Reading and recording	All 16 readings and units correct	Some readings correct	Reading method needs correction
Inference	Both comparisons use data and limits	Correct trend; incomplete evidence	Does not match data
Communication	Four speeds, both axes and four points correct; D separate	Some calculations / points correct	Method needs correction
Concepts	Links light, transpiration pull (蒸騰拉力) and xylem (木質部) transport; distinguishes processes	Some mechanism correct	Confuses transport and transpiration (蒸騰)
Checking and correction	Identifies and reasonably corrects original error	Incomplete reason	Copies key only / keeps error

Using the complete answer key

C1 contains observation, hypothesis evaluation, variables, assumptions, setups, all readings, speeds and choice answers. C2 contains the graph, concept explanation and correction examples. Diagnose original B responses before reviewing C corrections; copying answers is not understanding.

This guide is a draft for piloting and rater calibration, not a validated instrument.

For teachers only. Do not give this part to students.

Question alignment • 1–5

VL original	Treatment	Paper task	Shared goal	Reason
Your first observations	Retain	B1 Describe visible changes in the trace; keep explanations separate from observations.	Observation	Separate observation and inference
If light intensity (光強度) increases, the upward speed of the red water trace will (increase / decrease / show no clear change).	Adapt	A2 → B1 Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.	Testable (可測試的) hypotheses	Teach, then evaluate; not original prediction
My reason	Adapt	B1 Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.	Explain testability	State factor and measured outcome
Identify the variables	Retain classification	A2 → B1 For the light comparison using A, B and C, tick one column for each factor.	Classify variables	Same seven factors and three classes
What assumptions does this investigation make?	Retain	B2 What assumptions does this investigation make? Changing lamp distance may change temperature. Why monitor and keep all setups at 25°C?	Assumptions and fairness	Keep choices and explain temperature control

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Question alignment • 6–10

VL original	Treatment	Paper task	Shared goal	Reason
How would you plan a comparison setup to find out whether leaves help with water transport?	Equivalent adaptation	B2 For light intensity (光強度) use (A/B/C / A/D); for leaf presence use (B/D / A/D). Why is A/D not a fair comparison of leaf presence?	Fair comparisons and controls	Identify comparisons in specified setups
Include the celery, leaves, red water, lamps, distances and measurement positions. Put 2 cm of each celery leaf stalk into the red water. Setup labels / written design	Complete labels	B2 ① points to the long stalk, ② to the liquid surface, ③ to the immersed part. Complete: plant part and water-carrying tissue for ①; ruler reading at ②; immersion depth for ③.	Understand setups and conditions	Read and justify controls instead of free design
All four setups run together. Take measurements every 10 minutes of simulated time.	Static	B2–B4 Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.	Identify ruler procedure	Not full practical operation
Your measurement record (0/10/20/30 min)	Static ruler reading	A2 → B2–B4 Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.	Read and record	Same four times; neutral example first
Calculate the average upward speed	Retain	A2 → B4 Average upward speed = (height at 30 min − height at 0 min) ÷ 30. Use two decimal places and cm/min.	Calculate mean speed	Same formula and two decimals

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Question alignment • 11–13

VL original	Treatment	Paper task	Shared goal	Reason
Light intensity (光強度) and the upward speed of the red water trace	Paper plotting	A2 → B5 Plot four points using your calculations. X: light intensity (光強度) (lux); Y: average upward speed (cm/min). Join A, B and C in light-intensity order to show a reasonable trend. Mark D separately; do not join it into that trend.	Graph communication	Same axes; D separate
As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend).	Retain with evidence	B5 As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend). Use data from A, B, C and B/D to support the conclusions.	Light trend	Same choices and data
At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar). This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards).	Retain	B5 At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar). This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards). Use data from A, B, C and B/D to support the conclusions.	B/D comparison and leaf role	Includes original leafConclusion item

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Question alignment • 14–15

VL original	Treatment	Paper task	Shared goal	Reason
We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).)	Retain with limits	B5 → C1 We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).) Can three light-intensity points give exact speeds for all untested values? (yes / no)	Evidence limits	Transport is not transpiration (蒸騰); no exact extrapolation
Do your data support your original hypothesis? Use ideas from the key learning points above to improve your original explanation. Explain how it relates to your data.	Adapt	C2 Use A/B/C and B/D data to explain light, transpiration (蒸騰) and xylem (木質部) transport, including suitable conditions and evidence limits. Choose a wrong or incomplete answer of your own: state it and correct it with data or concepts. If none, correct “trace speed is transpiration rate (蒸騰速率)”.	Explain and correct	Use data and correct mistakes

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